Aly Dhedhi

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https://dhedhialy.github.io/

EDUCATION

American Heritage Schools - Broward Campus
High School Diploma(Premed Program) **Plantation, FL**
June 2026–June 2029

- Weighted GPA ~ 4.0/4.0, Unweighted GPA ~ 4.0/4.0
- AP Biology, AP Precalc, APUSH, Pre-med Terminologies, Honours Chemistry, Honours Biology
- Involved in: HOSA, Pre-Med Society, USABO, Mu Alpha Theta,

Jumeirah College
High School Diploma **Dubai, UAE**
August 2024–May 2026

- Weighted GPA ~ 3.9/ 4.0, Unweighted GPA ~ 3.8/4.0
- Pursued GCSE qualifications in core subjects (Math, English Literature and Language, Sciences) along with: Economics, Geography, Food Science and Design Technology.
- Involved in: MUN, WSC, JC Econ magazine; Founder of Mu Alpha Theta's JC chapter
- Captain of JC Rugby Squad (2024-25)

WORK & LEADERSHIP EXPERIENCE

Emergence Forecasting Model for Protein Hotspots **Cambridge, MA**
(Kellis Lab)(In Person)(Research) May 2026–Present

- Built a predictive machine learning pipeline using XGBoost to forecast unobserved drug-resistant mutations in *M. tuberculosis* with a 0.971 AUROC.
- Integrated population homoplasy data with AlphaFold structural features and drug-pocket spatial coordinates to rank 44,016 accessible variants.
- Validated model against 12,287 CRyPTIC clinical isolates and performed molecular docking to discover a literature-novel candidate mechanism (*gyrB* Q538L).
- First-author paper in preparation for pre-print submission to arXiv.
- Planning a post-publication extension, beyond tuberculosis, specifically, **Alzheimer's**, and **MRSA**
- Paper: tb-resistance-discovery
- Github repo: https://github.com/DHEDHiAly/tb-resistance-discovery

British Physics Olympiad **Dubai, UAE**
(Gold Award) March 2026

- Placed in the top tier of ~9,000 students worldwide, demonstrating advanced problem-solving and critical thinking skills in unfamiliar, undergraduate-style theoretical and numerical scenarios.

Clinical Bias Research in Clinical Guidelines **Cambridge, MA,USA**
(MIT Critical Data) March 2026–Present

- Developed a pipeline that analyzes clinical guideline DOIs, identifies cited studies, and extracts participant demographics to assess evidence representation.([RADAR](#))
- Built automated workflows for citation retrieval, demographic extraction, and evidence-base visualization across major clinical guidelines.

Hospital-Invariant ICU Representation Learning & Counterfactual Modeling **Dubai, UAE**
(Research) March 2026–April 2026

- Built a MIMIC-IV mortality prediction pipeline using K-Means phenotype discovery and gradient-boosted learning.
- Improved predictive performance through cluster-augmented representations, achieving AUROC 0.869 and AUPRC 0.302.
- First-author paper accepted to the IEEE Symposium on Computational Intelligence in Bioinformatics and Computational Biology 2026.
- Paper: ICU Mortality Final.pdf
- Github repo:ICU RESEARCH

Computational Peptide Design for BBB-Penetrant EGFRvIII Targeting **Dubai, UAE**
(Research) November 2025–March 2026

- Built an automated in silico pipeline for BBB-penetrant peptide design targeting EGFRvIII in glioblastoma.
- Identified a top-ranked cyclized Angiopep-2/GE11 fusion candidate through structural prediction, docking, and interface analysis.
- First-author paper accepted to the IEEE Symposium on Computational Intelligence in Bioinformatics and Computational Biology 2026.
- Paper: Designing_a_BBB_Penetrant_Peptide_Final (1).pdf
- Github repo:BBB RESEARCH

Computer Science and Artificial Intelligence Laboratory (MIT) **Cambridge, MA, USA**
Full-Stack Software Engineer(Hybrid) November 2025 –Present

- Leverage AI and representation learning to model high-dimensional clinical data, constructing unified patient embeddings and longitudinal trajectory representations from MIMIC-IV.
- Design and implement a two-stage pipeline combining multimodal embedding (scaled labs + diagnosis text embeddings) with clustering (KMeans) to derive latent clinical states and patient-level temporal trajectories.
- Engineer trajectory-level features (state transitions, path length, temporal duration, cluster displacement) to quantify disease progression dynamics and enable secondary clustering of patient progression patterns.
- Analyze and visualize large-scale patient transitions (~millions of temporal steps) through embedding spaces, identifying structurally distinct progression phenotypes across cohorts.
- Validate structure and robustness using dimensionality reduction (PCA/UMAP), cluster stability metrics, and transition pattern analysis, producing reproducible artifacts for downstream modeling.
- Mentored under Manolis Kellis and an interdisciplinary team

The Citizens Foundation(TCF) **Dubai, UAE**
Student Fundraiser and Volunteer July 2022 – Present
- Raised over 23,000$ to support underprivileged children's education; equivalent to a year's worth of education for 166 children
- Invited to deliver speeches at national fundraising events to advocate for education access and encourage donor participation
- Coordinated fundraising campaigns and engaged peers to maximize outreach and contributions; Recognised at national dinners as the top fundraiser for the youth cohort twice (December 2023, December 2025)

HCA Florida – West Palm Beach **West Palm Beach, FL**
Gastroenterologist Assistant / Clinical Shadowing Student *July 2025 – August 2025*

- Assisted a gastroenterologist: Dr Adam D. MD in patient consultations, observing diagnostic workflows and treatment planning processes
- Gained hands-on exposure to clinical procedures, including endoscopic techniques, biopsy collection, and patient monitoring
- Developed understanding of gastroenterology practice, patient care, and interdisciplinary collaboration in a hospital setting

Boy Scouts of America, Troop 813 **Dubai, UAE**
Troop Guide / Assistant SeniorPatrol Leader *August 2021 – Present*

- Current Life Scout; Eagle Scout expected by mid 2027
- Led patrols, organized community service projects, and mentored younger scouts in leadership and civic responsibility
- Key merit badges earned: Lifesaving, First Aid, Shotgun Shooting, Rifle Shooting, Citizenship in the Nation, World, and Community, Communication, Sustainability, Golf,

ADDITIONAL SKILLS AND INTERESTS

Licenses and Certifications: CITI Data or Specimens Only Research; CITI Conflict of Interest; Trinity College London Piano (Grade 5),
Technical:
Python (intermediate; scientific computing, data pipelines), PyTorch (foundational), scikit-learn, NumPy, pandas, Matplotlib; BioPython (sequence analysis, physicochemical profiling); protein structure tools/APIs (ESMFold); molecular docking workflows (PatchDock); Google Colab (cloud-native pipelines); HTML5 (fluent); Figma (UI/UX design, prototyping); Microsoft Office Suite, Google Workspace; experience with Manus, MGX, Bolt AI, Lovable, Base44

Laboratory & Research:
Computational molecular biology and peptide design; in silico structural modeling and docking-based screening; physicochemical characterization and BBB permeability heuristics; experimental design and reproducible pipeline development; literature-driven hypothesis formulation and scientific writing

Languages: English (fluent), Urdu (fluent), Arabic (intermediate), Mandarin (intermediate)

Interests: Biomedical research, AI in healthcare, clinical phenotyping, oncology, Muay Thai, tennis, golf, fundraising initiatives